ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 1ST YEAR

DISCRETE MATHEMATICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 1st year Discrete Mathematics paper 1 · practice a.

Written by Academic Central for practice, following first-year Computer Science and Mathematics syllabuses at South African universities (discrete mathematics for computer science). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct steps even if a later step is wrong (consistent accuracy). Accept any correct proof or method that differs from the one shown, and equivalent notation.

QUESTION 1

1.1.1 $p \rightarrow q$ ✓ (1)

1.1.2 $q \rightarrow p$ ✓ (1)

1.1.3 $\neg p \wedge \neg q$ ✓ (1)

1.2 (5)

p	q	¬p	¬q	p → q	¬q → ¬p	(p → q) ↔ (¬q → ¬p)
T	T	F	F	T	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

¬p and ¬q columns ✓ p → q column ✓ ¬q → ¬p column ✓ final column ✓
Yes, it is a tautology because the final column is true in every row ✓

1.3 $\neg(p \vee \neg q) \vee (\neg p \wedge \neg q)$ (4)
$\equiv (\neg p \wedge q) \vee (\neg p \wedge \neg q)$ De Morgan's law and double negation ✓
$\equiv \neg p \wedge (q \vee \neg q)$ distributive law ✓
$\equiv \neg p \wedge T$ negation (complement) law ✓
$\equiv \neg p$ identity law ✓

1.4 Converse: If n^2 is even, then n is even ✓ (3)
Inverse: If n is not even (odd), then n^2 is not even (odd) ✓
Contrapositive: If n^2 is not even (odd), then n is not even (odd) ✓

1.5.1 TRUE: the square of any integer is zero or positive ✓ (1)

1.5.2 FALSE: $\sqrt{2}$ is not an integer, so no integer squares to 2 ✓ (1)

1.5.3 TRUE: for every x choose $y = -x$ ✓ (1)

1.6 $\neg\forall x\, \exists y\, (x < y) \equiv \exists x\, \forall y\, \neg(x < y)$ ✓ $\equiv \exists x\, \forall y\, (x \geq y)$ ✓ (2)

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QUESTION 2

2.1.1 $\{2, 4\}$ ✓ (1)

2.1.2 $\{1, 2, 3, 4, 5, 6, 9\}$ ✓ (1)

2.1.3 $\{6, 8, 10\}$ ✓ (1)

2.1.4 $A \cup B = \{1, 2, 3, 4, 5, 6, 8, 10\}$, so $(A \cup B)' = \{7, 9\}$ ✓ (1)

2.1.5 $5 \times 3 = 15$ ✓ (1)

2.2 $2^3 = 8$ ✓ $\{3, 6\}, \{3, 9\}, \{6, 9\}$ ✓ (2)

2.3 ($\subseteq$) Let $x \in A - (B \cup C)$. Then $x \in A$ and $x \notin B \cup C$, so $x \notin B$ and $x \notin C$ ✓. Therefore $x \in A - B$ and $x \in A - C$, so $x \in (A - B) \cap (A - C)$ ✓ (4)
($\supseteq$) Let $x \in (A - B) \cap (A - C)$. Then $x \in A$, $x \notin B$ and $x \notin C$ ✓, so $x \notin B \cup C$ and $x \in A - (B \cup C)$ ✓
Both inclusions hold, so the sets are equal.

2.4.1 $n(M \cup P) = 70 + 50 - 30 = 90$ ✓ Neither $= 120 - 90 = 30$ ✓ (2)

2.4.2 Mathematics only $= 70 - 30 = 40$ ✓ Physics only $= 50 - 30 = 20$ ✓ (2)

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QUESTION 3

3.1 Reflexive: yes, $(a, a) \in R$ for every $a \in A$ ✓ (4)
Symmetric: yes, for every (a, b) in R, (b, a) is also in R ✓
Antisymmetric: no, $(1, 2)$ and $(2, 1)$ are both in R but $1 \neq 2$ ✓
Transitive: yes, e.g. $(1, 2)$, $(2, 1)$ give $(1, 1) \in R$; every such pair has its result in R ✓

3.2 Yes, it is reflexive, symmetric and transitive ✓ (3)
Classes: $[1] = \{1, 2\}$ ✓ $[3] = \{3, 4\}$ ✓

3.3.1 $f(2) = 1$, $g(1) = 1^2 + 1 = 2$ ✓ (1)

3.3.2 $f(g(x)) = 3(x^2 + 1) - 5$ ✓ $= 3x^2 - 2$ ✓ (2)

3.3.3 One-to-one: if $3a - 5 = 3b - 5$ then $a = b$ ✓ (3)
Onto: for any $y \in \mathbb{R}$, $x = (y + 5)/3$ gives $f(x) = y$ ✓
$f^{-1}(x) = (x + 5)/3$ ✓

3.4 No ✓: $g(1) = g(-1) = 2$ but $1 \neq -1$ ✓ (2)

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QUESTION 4

4.1 Let n be odd, so $n = 2k + 1$ for some integer k ✓ (4)
$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1$ ✓ $= 2(2k^2 + 2k) + 1$ ✓
$2k^2 + 2k$ is an integer, so n^2 is odd ✓

4.2 Contrapositive: if n is even, then $3n + 2$ is even ✓ (4)
Let $n = 2k$ for some integer k ✓
$3n + 2 = 6k + 2 = 2(3k + 1)$ ✓, which is even. The contrapositive is true, so the statement is true ✓

4.3 Base case $n = 1$: left side $= 1$, right side $= 1^2 = 1$, so it is true ✓ (7)
Assume it is true for $n = k$: $1 + 3 + \ldots + (2k - 1) = k^2$ ✓
For $n = k + 1$: $1 + 3 + \ldots + (2k - 1) + (2(k + 1) - 1)$ ✓
$= k^2 + 2k + 1$ (using the assumption) ✓ ✓
$= (k + 1)^2$ ✓
So if it is true for k, it is true for $k + 1$. By the principle of mathematical induction it is true for all $n \geq 1$ ✓

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QUESTION 5

5.1.1 $26 \times 26 \times 10 \times 10 \times 10 \times 10$ ✓ $= 6\,760\,000$ ✓ (2)

5.1.2 $26 \times 26 \times 10 \times 9 \times 8 \times 7$ ✓ $= 3\,407\,040$ ✓ (2)

5.2.1 $C(12, 4)$ ✓ $= 495$ ✓ (2)

5.2.2 $C(7, 2) \times C(5, 2) = 21 \times 10$ ✓ $= 210$ ✓ (2)

5.2.3 All committees minus all-women committees: $495 - C(7, 4) = 495 - 35$ ✓ $= 460$ ✓ (2)

5.3 $6! / (2! \times 2!)$ (two T's and two E's) ✓ $= 180$ ✓ (2)

5.4 There are 12 months (pigeonholes) and 13 people (pigeons) ✓. Since $13 > 12$, at least one month must have two or more people ✓ (2)

5.5 $C(7, 3) = 35$ ✓ (1)

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QUESTION 6

6.1 5 vertices ✓ 6 edges ✓ (2)

6.2 deg(A) = 2, deg(B) = 3, deg(C) = 3, deg(D) = 3, deg(E) = 1 ✓✓ (one mark off per error) (3)
Sum of degrees = $12 = 2 \times 6$ edges ✓

6.3 G has four vertices of odd degree (B, C, D and E) ✓ (3)
An Euler circuit needs all degrees even, so there is no Euler circuit ✓
An Euler path needs exactly 0 or 2 odd vertices, so there is no Euler path ✓

6.4 (3)

	A	B	C	D	E
A	0	1	1	0	0
B	1	0	1	1	0
C	1	1	0	1	0
D	0	1	1	0	1
E	0	0	0	1	0

✓✓✓ (one mark off per wrong row, minimum 0)

6.5 No ✓: it contains a cycle, e.g. A–B–C–A (a tree with 5 vertices has only 4 edges) ✓ (2)

6.6 (5)

Step (vertex fixed)	S	A	B	C	D	T
S	0	4	2	∞	∞	∞
B		3	2	10	12	∞
A		3		8	12	∞
C				8	10	14
D					10	13
T						13

Correct updates: A becomes 3 via B ✓; C becomes 8 via A ✓; D becomes 10 and T becomes 13 ✓
Shortest path S → B → A → C → D → T ✓ Length 13 ✓

6.7 $n(n-1)/2 = 6 \times 5 / 2 = 15$ ✓✓ (2)

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TOTAL: 100

END OF MARKING GUIDELINE